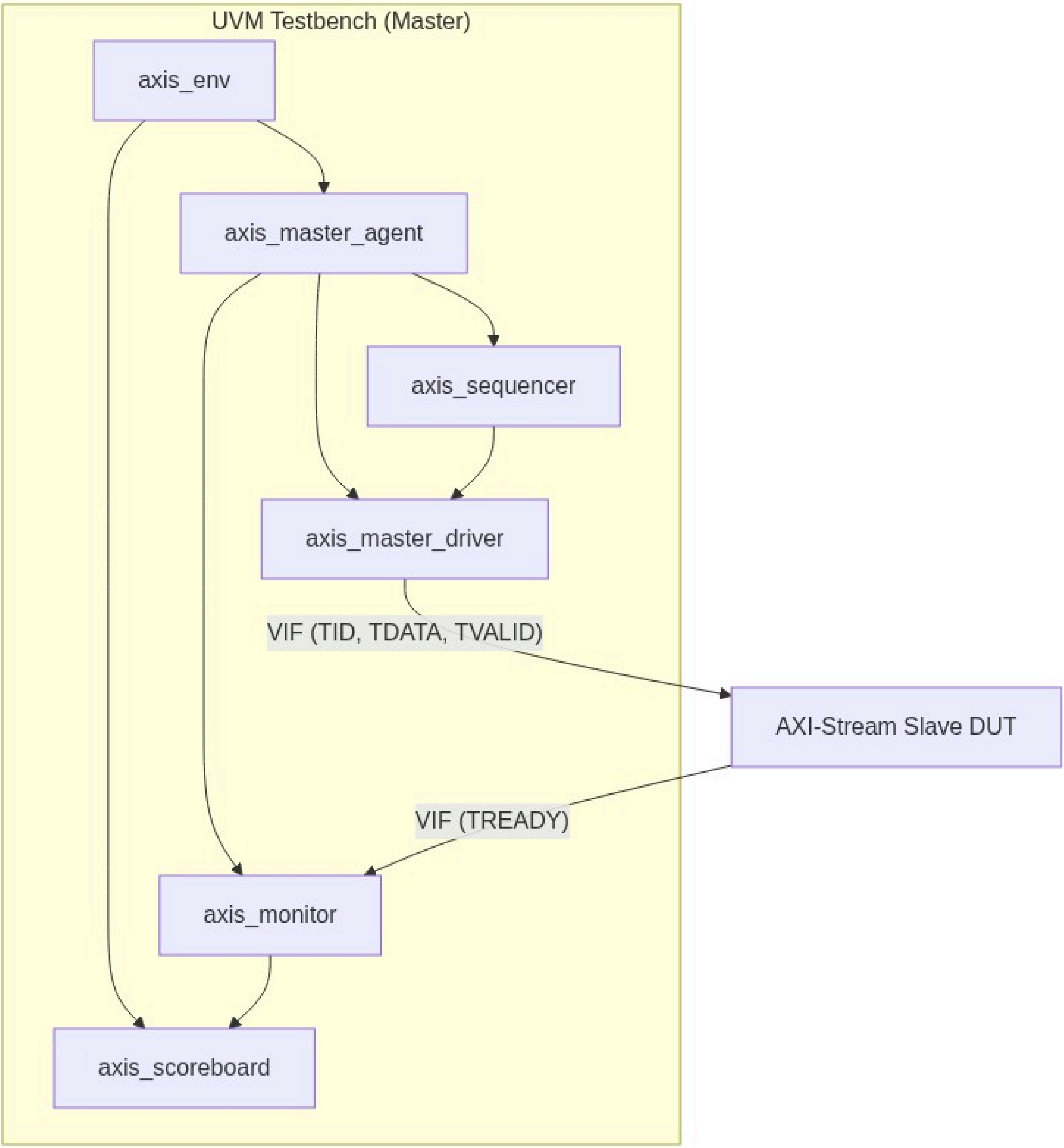
Verification Plan Report

Project: AXI-Stream Full-Stack Verification

Version: 1.8

Verification Role: Master

1. UVM Environment Architecture



Verification Plan Report

2. Test Requirements

Index	Feature/Sub	Name	Description
TR_01	Handshake (Stability)	Master Stability	Master must keep TDATA stable if TVALID=1 and TREADY=0.
TR_02	Packet (ID)	TID Consistency	TID must not change during a packet.

3. Test Cases & Mapping

Index	Scenario	TR Mapping
TC_01	Protocol Stability: Strobe TVALID and hold TREADY LOW. Verify TDATA remains frozen.	TR_01
TC_02	Packet Boundary Stress: Send varied packet sizes (1 to 256) with randomized TID.	TR_02

4. Functional Coverage Groups

Group Name	Description
cg_axis_handshake	Coverage of TVALID/TREADY handshakes.

5. Interface Signals

Signal Name	Direction	Width	Description
TDATA	Master	TD_W	Data payload.
TVALID	Master	1	Valid indicator.
TREADY	Slave	1	Back-pressure.
TLAST	Master	1	End of packet.

6. Verification Environment Structure

`verif/axis_master_env/`

```
|-- agents/
|   |-- axis_master_agent.sv
|   |-- axis_master_driver.sv
|   |-- axis_monitor.sv
|   |-- axis_sequencer.sv
|-- env/
|   |-- axis_env.sv
```

```
|   |-- axis_scoreboard.sv
|   |-- axis_env_pkg.sv
|-- sequences/
|   |-- axis_base_seq.sv
|   |-- axis_random_seq.sv
|   |-- axis_packet_seq.sv
|-- tests/
|   |-- axis_base_test.sv
|   |-- axis_stress_test.sv
|-- top/
    |-- axis_tb_top.sv
    |-- axis_if.sv
```

7. SystemVerilog Implementation

AXI-Stream Master Monitor - Protocol Check

```
task axis_monitor::collect_transfer();
  forever begin
    @(posedge vif.ACLK);
    if (vif.ARESETn && vif.TVALID && vif.TREADY) begin
      axis_transaction tr = axis_transaction::type_id::create("tr");
      tr.data = vif.TDATA;
      tr.last = vif.TLAST;
      tr.id   = vif.TID;
      ap.write(tr);
    end
  end
endtask
```

Handshake Stability Assertion

```
property p_tvalid_stability;
  @(posedge ACLK) disable iff (!ARESETn)
  (TVALID && !TREADY) |=> $stable(TDATA) && $stable(TLAST);
endproperty
assert_tvalid_stability: assert property(p_tvalid_stability);
```